

Reserve-oir Dogs: The Great Stablecoin Standoff

Thin Cushions and False Security: Evaluating Stablecoin Resilience under the GENIUS Act

Abstract Stablecoins underpin settlement and funding in cryptocurrency spot markets, yet their pegs can fail when redemption credibility is questioned. We use the March 2023 SVB episode as an exogenous shock to study how issuer fundamentals shape peg stability, market liquidity, and cross-market stress transmission for USD stablecoins. We combine a top-down analysis of market dislocations and spillovers with a bottom-up reconstruction of issuer reserves to link peg deviations to balance-sheet conditions and to conduct banking-style stress tests. We find that reserve-quality and disclosure requirements improve baseline stability, but do not substitute for loss-absorbing capital in run-like scenarios. Issuers can meet high-quality reserve standards and still remain vulnerable to rapid redemptions without explicit equity buffers. These results suggest that the 2025 GENIUS Act could strengthen the framework by pairing reserve mandates with minimum capital requirements.



I. INTRODUCTION

Fiat-backed stablecoins are digital assets designed to maintain parity with a reference currency. USD Coin (USDC), issued by Circle, and Tether (USDT), issued by Tether, are designed to trade at par with the U.S. dollar (USD). Between December 2021 and February 2026, USDC grew by approximately 74% to a market capitalization of \$73.9 billion, while USDT expanded by roughly 135% to \$183.7 billion,¹ establishing both instruments as primary funding and settlement rails (channels) for digital asset markets and enabling assets such as Bitcoin to trade simultaneously against USD, USDT, and USDC across fragmented exchanges.

Stablecoin stability depends on the credibility, liquidity, and composition of issuer reserves, as well as market structure and arbitrage conditions. When redemption pressure intensifies, issuers may need to liquidate assets under adverse conditions, generating fire-sale externalities that feed back into secondary-market pricing. During the March 2023 SVB episode, basis spreads that normally sat within 10 bps blew out beyond 900 bps for USDC-quoted pairs, while USDT-quoted pairs remained largely unaffected [4]: two instruments designed to be dollar equivalents, traded at materially different prices on the same exchanges at the same moment.

The proximate cause was Circle’s disclosure that \$3.3 billion of USDC reserves were held at SVB and temporarily inaccessible following its closure, triggering an immediate reassessment of USDC redeemability and a flight toward USDT as the perceived safer settlement asset. Yet USDT is the offshore, less transparent, and by conventional metrics, the riskier of the two instruments. This inversion raises a question the existing literature has not fully pinned down: whether reserve quality alone is sufficient to sustain peg credibility under run-like

conditions, or whether loss-absorbing capital plays a distinct and irreplaceable role.

This brings us to consider the 2025 GENIUS Act² and assess which vulnerabilities its provisions address and which it does not. USDC is broadly aligned with the Act’s reserve and transparency standards, whereas USDT remains largely outside the U.S. regulatory perimeter due to its offshore operations. This divergence allows us to ask whether reserve-based requirements are sufficient to contain the redeemability-risk channel our analysis documents, or whether explicit minimum equity requirements analogous to Tier 1 capital in banking are a necessary complement.

To answer this, we take two complementary approaches. First, a top-down market view quantifies how stress propagated across exchanges and quote-currency venues through basis dislocations, liquidity deterioration, and volatility spillovers between stablecoins and traditional asset classes. Second, a bottom-up issuer view reconstructs daily reserve portfolios for Circle and Tether to estimate panel models linking peg deviations to balance-sheet metrics and to apply banking-style stress tests to each issuer’s asset-liability structure.

These approaches are united by a main (H_1) and supporting hypothesis (H_2):

H_1 : Reserve quality alone is not enough to keep peg credibility in run-like conditions. Loss-absorbing capital has a unique and irreplaceable role that reserve mandates cannot replace.

H_2 : USDT and USDC are structurally different instruments: USDT is an offshore liquidity vehicle with a capital buffer, while USDC is TradFi-integrated, i.e., it is reserve quality compliant but with a thin cushion. This structural difference, not the quality of the reserves, explains the asymmetric market response during the SVB episode.

¹Information from CoinGecko’s market capitalization data. From December 31, 2021, to February 20, 2026.

²Guiding and Establishing National Innovation for U.S. Stablecoins Act.

This paper is structured as follows. Section II documents the SVB shock and its market impact across exchanges and quote currencies. Section III summarizes the GENIUS Act’s scope, key provisions, and implementation timeline, and then presents our dual empirical analysis: (i) stress transmission and volatility spillovers, and (ii) issuer-level resilience based on reconstructed daily reserves, panel models of peg dynamics, and banking-style stress tests. Section IV concludes by evaluating what the GENIUS framework achieves and what implementing regulations may still need to be addressed ahead of the January 2027 compliance deadline.

II. ILLIQUIDITY, FRAGMENTATION & THE LIMITS OF SELF-REGULATION

On March 10, 2023, USDC’s order books, arbitrage channels, and peg stability broke down simultaneously, not uniformly, across stablecoins.³ We analyze this breakdown across three complementary lenses.

Cross Exchange Arbitrage Gaps

Cross-exchange arbitrage gaps measure the price dispersion of the same asset pair across different exchanges, expressed in basis points. Larger gaps indicate market fragmentation and reduced integration, reflecting periods when arbitrage capital cannot fully enforce price convergence, consistent with standard cross-venue pricing frameworks [4].

$$Arb\ Gap_t = \left| \frac{P_{Exch1,t} - P_{Exch2,t}}{\frac{P_{Exch1,t} + P_{Exch2,t}}{2}} \right| \times 10,000 \quad (1)$$

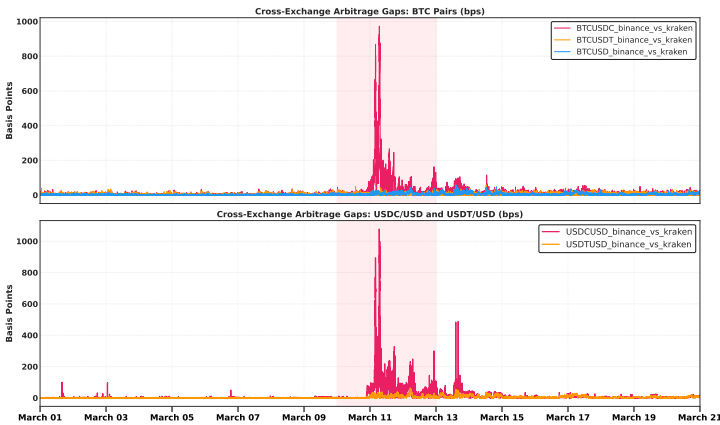


Fig. 1: Cross Exchange Arbitrage Gaps across Exchanges

³We collected 1-minute OHLCV data via exchange REST APIs for Binance.US and Kraken across BTC/USD, BTC/USDC, BTC/USDT, USDC/USD, and USDT/USD pairs. Short gaps (≤ 5 minutes) are filled via LOCF; medium gaps (6-30 minutes) via ETS trained exclusively on pre-gap data; gaps exceeding 30 minutes are dropped to prevent model interpolation over structurally missing periods. Additionally, we collected Level 2 order book data for BTC/USD and BTC/USDT in Coinbase using CoinAPI.

Pre-crisis, gaps across all pairs remained within 0-10 bps ($\leq 0.1\%$), indicating tightly integrated markets. Following SVB’s collapse, fragmentation became extreme and concentrated in USDC: BTC/USDC gaps surged to ~ 900 -1,000 bps (≈ 9 -10%) and USDC/USD exceeded 1,000 bps ($>10\%$), while BTC/USDT and BTC/USD remained below 100 bps ($<1\%$) - a ten-fold divergence on the same exchanges. As shown in Fig. 1, asymmetric widening reflects reserve uncertainty after Circle disclosed SVB exposure: arbitrage capital withdrew from USDC specifically, raising execution risk in USDC-denominated markets and revealing that the fragmentation was issuer-specific, not systemic to crypto.

Market Quality Index (MQI)

We construct the MQI by combining four standard microstructure components [8], [1]: the Corwin–Schultz bid–ask spread (tightness) [10], the Amihud illiquidity ratio (price impact per dollar traded), VPIN (order-flow toxicity)⁴, and an efficiency stress measure (disruptions in price discovery).

$$MQI_t = 100 \left[\begin{aligned} &w_1(1 - \widetilde{Spread}_t) + w_2(1 - \widetilde{Illiq}_t) \\ &+ w_3(1 - \widetilde{EffStress}_t) + w_4(1 - \widetilde{VPIN}_t) \end{aligned} \right] \quad (2)$$

where $w_1 = w_2 = 0.30$ and $w_3 = w_4 = 0.20$. Each component is standardized against a pre-crisis baseline (March 1–9, 2023), normalized to $[0, 1]$, and inverted so that higher values indicate stronger market functioning (0 = severe breakdown, 100 = ideal). The components are:

$$Spread_t = 2(e^{\alpha_t} - 1) \quad (3)$$

$$ILLIQ_t = \frac{1}{I} \sum_{i=1}^I \frac{|r_i|}{P_i V_i} \times 10^6 \quad (4)$$

$$VPIN_i = \frac{|V_i^B - V_i^S|}{V_i} \quad (5)$$

$$EffStress_t = \frac{|Eff_t - \mu_{pre}|}{\sigma_{pre}} \quad (6)$$

An analogous **Stablecoin Peg Quality Index (SPQI)** applies the same framework to USDC/USD and USDT/USD pairs, where values near 100 indicate orderly peg trading, and sharp declines indicate signal peg stress.

From Fig. 2, we observe that during Pre-crisis (March 1-9), BTC/USD, BTC/USDC, and BTC/USDT cluster at MQI levels of 70–85, confirming functional parity across pairs. The SVB failure fractured this immediately: BTC/USDC collapsed 30-40 points to 30-45 while

⁴Order-flow toxicity represents the risk that liquidity providers are being adversely selected by better-informed traders.

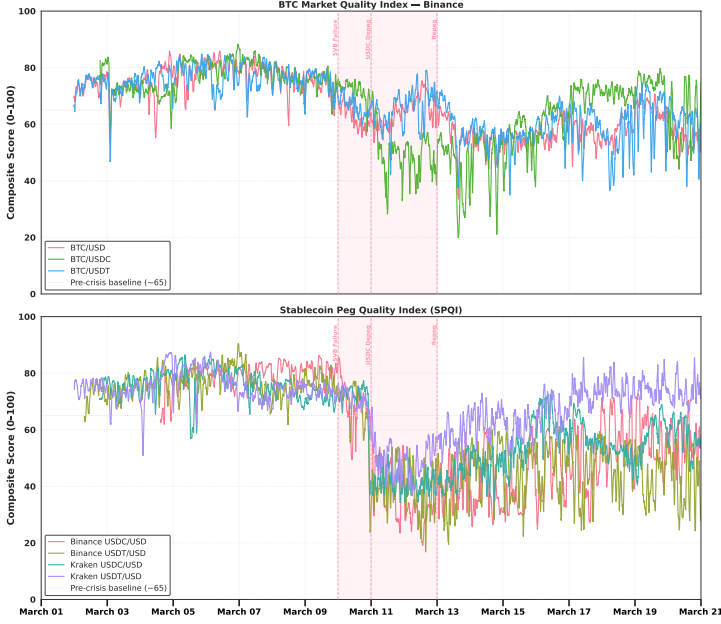


Fig. 2: Quality Index Analysis

BTC/USDT held at 50-60 — a currency-specific shock, not a systemic BTC event.

The SPQI panel above corroborates that the SVB failure was a currency-specific shock rather than a systemic BTC event across both exchanges, ruling out any venue-specific artifact. The nominal peg recovered on March 13; functional liquidity did not - BTC/USDC liquidity remained suppressed through March 21, meaning markets continued pricing residual reserve risk long after Circle’s reassurance. In the next paragraph, we apply Kyle’s λ to show why this occurred at the order-book level.

Rolling Kyle’s Lambda

To quantify the price impact of market depth, we ran regressions based on the framework established by Kyle [2]. The standard regression is written as:

$$\Delta P_t = \alpha + \lambda \cdot S_t + \epsilon_t \quad (7)$$

where ΔP_t is the mid-price change and S_t is the volume of signed trade.⁵ Higher λ values represent liquidity droughts and imply that trades have a large impact on price.

On Coinbase, BTC is traded largely through the BTC/USD pair—equivalent to BTC/USDC given Coinbase’s 1:1 USD–USDC treatment—while the BTC/USDT pair maintains a much thinner order book. During the SVB crisis, the upward shock in λ values (Fig. 3) for BTC/USD indicates that liquidity briefly evaporated, with the order book becoming as thin as the

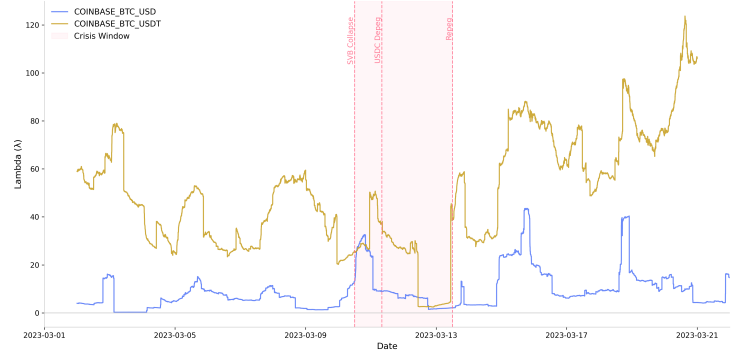


Fig. 3: Rolling Kyle’s Lambda for BTCUSD vs BTCUSDT

chronically illiquid BTC/USDT pair, and price impact per unit of volume increased significantly. In contrast, the temporarily declining λ values for BTC/USDT imply that the de-pegging event thickened its order book. The crossing of the two λ series captures this substitution dynamic directly: uncertainty around reserve liquidity, fragmented price convergence, market quality, and order-book depth, draining liquidity from USDC-linked markets and reallocating it into USDT.

The remainder of this paper tests whether the GENIUS Act alters this structural response.

III. “GENIUS” ACT: TO STABILISE STABLECOINS

The SVB episode showed that stablecoin pegs are vulnerable to rapid, confidence-driven runs with spillovers into broader crypto market dislocations. In response, the U.S. enacted the GENIUS Act in July 2025, establishing a federal framework for significant stablecoin issuers (e.g., > \$10bn) centered on 1:1 reserve backing in high-quality liquid assets, enhanced disclosure, and legal clarity by defining payment stablecoins as digital assets distinct from securities, commodities, or currencies.⁶ The Act also prohibits issuers from paying interest or yield on stablecoin balances, reinforcing separation from interest-bearing deposits and limiting incentives for deposit-like flight from the banking system.⁷ Finally, it formalizes eligible issuers, opening issuance to both insured depository institutions and federally licensed nonbanks under uniform reserve and compliance standards.

These provisions meaningfully improve transparency and baseline reserve quality, but the SVB episode highlights what they do not guarantee. Circle’s reserves were already largely consistent with GENIUS-style HQLA standards in March 2023, yet USDC de-pegged after the \$3.3bn reserves in SVB became inaccessible.

⁶Eligible reserves include U.S. dollars, short-term Treasuries, Treasury-backed repo, and other regulator-approved low-risk government instruments.

⁷The Act does not prohibit exchanges from offering yield to stablecoin holders.

⁵Following [9], we substitute signed trade volume with Order Flow Imbalance (OFI), measuring net buying pressure at the top of the book using a Rolling OLS model with a rolling 1,440-minute window.

This was not a reserve-quality failure; it was market panic combined with liquidity placement failure that a loss-absorbing capital buffer would be better positioned to mitigate under run-like conditions. This motivates two empirical exercises: (i) Top-down volatility spillover analysis testing whether GENIUS-aligned design translates into different stress transmission, and (ii) Bottom-up banking-style stress test implemented using our daily mark-to-market reconstruction of Circle and Tether reserves to assess whether GENIUS-compliant balance sheets can withstand a historically plausible 30% redemption shock (Circle aligned, Tether outside the perimeter).

A. From Safe Haven to Shock Transmitter: Market-Level Evidence

Market turmoil spreads across different markets to varying degrees depending on their interconnectedness. Information such as earnings announcements, geopolitical events, and macroeconomic data can trigger volatility in one market that transmits to others. **If spillovers are present, shocks in traditional markets should affect cryptocurrency-market volatility.**

During mild periods, market makers significantly contribute to market liquidity and trading volume. However, during financial crises or periods of market stress, market makers tend to reduce their activity due to the risk of adverse selection, which leads to a liquidity crunch. **We test if volatility in major non-stable cryptocurrencies (e.g., BTC, ETH) increases volatility in stablecoin exchange rates, with the effect amplified when market-maker liquidity provision declines.**

Furthermore, larger markets tend to be net transmitters of volatility to smaller ones, and negative shocks transmit volatility more strongly than positive ones. Stablecoins therefore act as a safe haven during stress events. **We test if USDC and USDT actually act as stores of value in cryptocurrency markets by using a Time Varying Parameter Vector Autoregressive Model (TVP-VAR).**

We observe that when stablecoins come under market stress, they become a net transmitter of volatility, whereas the forex market and USDT are net receivers of volatility.

Data: We aggregate daily OHLCV data using Binance.US and fill missing periods from Jun 9, 23 to Feb 19, 25⁸ using Coinbase and Kraken (USDC/USD only). Our variable set comprises eight daily series — USDT and USDC volatility as the dependent variables, and six external factors — Bitcoin, Ethereum, the S&P

500, the DXY dollar index, the KBW Bank Index, and gold (XAU). We also include the VIX level as a proxy for broad market fear rather than as a volatility measure, given its forward-looking nature. Since traditional markets do not trade on weekends, all series are sampled on weekdays only, yielding 1,603 observations from January 2020 to February 2026.⁹

The sample period spans from January 1, 2020 to February 20, 2026 including 1,604 trading days, which is sufficiently long to analyze volatility spillovers.

1) Descriptive Statistics: All of the above series are stationary (Augmented Dickey Fuller test), have fat tails, i.e., are not normal (Jarque Bera test) and are autocorrelated (Ljung Box(10)) at the 99% confidence interval indicating that we can use them in a regression without differencing and that there is a volatility clustering phenomenon in the dataset. USDC has a lower mean volatility (0.0023) than USDT (0.0039); any sudden change in USDC volatility during the SVB event is important because it is based on a calmer baseline. The KBW mean volatility (0.0127) is almost twice that of the S&P 500 (0.0075). This is in line with the fact that the banking sector was more unstable during the sample period, which is why it is included as a separate regressor.

2) What Predicts Stablecoin Volatility?: To examine potential drivers of deviation in stablecoin prices, we use contemporary values of these factors and lagged values of stablecoins to control for autocorrelation. We also control for year fixed effects, including 6 dummy variables corresponding to years 2020 to 2026:

$$V_{SC,t} = \beta_0 + \beta_1 V_{BTC,t} + \beta_2 V_{ETH,t} + \beta_3 V_{DXY,t} + \beta_4 V_{S\&P500,t} + \beta_5 V_{SC,t-1} + \beta_6 V_{KBW,t} + \beta_7 V_{XAU,t} + \beta_8 L_{VIX,t} + \beta_9^T \mathbf{D}_{year} + \varepsilon_t \quad (8)$$

(i) Ethereum volatility is the only crypto factor that significantly drives both stablecoins, suggesting Ethereum’s DeFi ecosystem is the primary crypto-channel through which shocks transmit.

(ii) The lagged own-volatility terms (0.57 for USDT, 0.58 for USDC) are dominant, confirming strong persistence and volatility clustering.

(iii) USDT’s VIX coefficient is negative and significant (0.0001**), while USDC’s is not — directly confirming the safe-haven interpretation. The cross-lagged term USDC Vol(t-1) transmitting to USDT (0.1359*) hints at a lead-lag relationship flowing from USDC to USDT.

As shown in Table I, both models are jointly significant ($p = 0.000$) and explain comparable shares of variation (USDT $R^2 = 0.46$, USDC $R^2 = 0.42$). DXY volatility

⁸SEC Lawsuit against Binance.US, suspension of fiat trading pairs

⁹All the traditional market data is collected using Bloomberg Terminal

is marginally significant for USDC (0.2059*) but not for USDT, consistent with USDC’s closer integration with the U.S. banking and FX systems. The remaining factors — BTC, S&P 500, KBW, and gold — are insignificant in both models, suggesting that equity market volatility and commodity price swings do not independently drive stablecoin volatility once ETH and the lagged terms are controlled for.¹⁰

TABLE I: Stablecoin Volatility Regressions

	(1) USDT	(2) USDC
Intercept	0.0011* (0.0007)	−0.0003 (0.0005)
BTC Volatility	−0.0035 (0.0032)	−0.0073 (0.0058)
ETH Volatility	0.0318*** (0.0072)	0.0291*** (0.0059)
DXY Volatility	0.0740 (0.1123)	0.2059* (0.1208)
S&P 500 Volatility	−0.0474 (0.0388)	−0.0495 (0.0639)
USDT Volatility _{t−1}	0.5700*** (0.0987)	0.0170 (0.0250)
USDC Volatility _{t−1}	0.1359* (0.0720)	0.5782*** (0.0910)
KBW Volatility	0.0414 (0.0260)	0.0721 (0.0584)
XAU Volatility	−0.0297 (0.0234)	0.0115 (0.0270)
VIX (Level)	−0.0001** (0.0000)	−0.0001 (0.0000)
Year dummies	Yes	Yes
Observations	1,602	1,602
R ²	0.4605	0.4227
Adj. R ²	0.4554	0.4172
F-statistic	32.6603	31.7401
Durbin–Watson	2.2188	2.1433
AIC	−11297.0372	−12012.5891
BIC	−11210.9731	−11926.5250

Notes: Newey–West HAC standard errors (maxlags = 10) in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The two instruments appear to play structurally distinct roles - unrelated to reserve quality - in the financial system. This divergence is supported by the asymmetric responses reported above, including USDT’s safe-haven behavior under high VIX and USDC’s sensitivity to banking-sector stress.

3) *Volatility Spillovers Across Markets:* The regression results establish which factors drive stablecoin volatility in isolation, but leave two questions unanswered. First, are stablecoins passive recipients of external shocks, or do they themselves transmit stress outward? Second, does the structural divergence between USDT and USDC (a safe-haven instrument versus a TradFi-integrated one) produce measurably different spillover profiles? To answer both questions, we estimate a Time-Varying Parameter Vector Autoregression (TVP-VAR) following [5].

The TVP-VAR system is estimated over the vector

$$X_t = [\text{USDT}_t, \text{USDC}_t, \text{BTC}_t, \text{ETH}_t, \text{S\&P500}_t, \text{DXY}_t, \text{XAU}_t, \text{KBW}_t]^\top$$

with time-varying coefficient matrices evolving as a random walk and time-varying covariance handled via

¹⁰Durbin–Watson statistics (USDT: 2.22, USDC: 2.14) confirm that the lagged terms adequately address serial correlation.

forgetting factors ($\lambda = 0.99$, $\kappa = 0.96$) following [6]. Lag length is set to $p = 2$ by BIC. To quantify directional transmission, we construct Generalised Forecast Error Variance Decompositions (GFEVD) from the vector moving-average representation of the system [7]:

$$\tilde{\lambda}_{ij}^g(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (\mathbf{e}_i^\top \mathbf{A}_h \Sigma \mathbf{e}_j)^2}{\sum_{h=0}^{H-1} (\mathbf{e}_i^\top \mathbf{A}_h \Sigma \mathbf{A}_h^\top \mathbf{e}_i)}, \quad (9)$$

where $\mathbf{A}_h$ are the moving-average coefficient matrices at horizon h , Σ is the residual covariance matrix, and $\mathbf{e}_i$ is a selection vector. Each row of $\tilde{\lambda}_{ij}^g(H)$ is normalised to sum to unity, so that the entry (i, j) measures the share of variable i ’s H -step-ahead forecast error variance attributable to shocks in variable j .

The Total Spillover Index (TSI) aggregates all off-diagonal entries. All measures are computed at $H = 10$ days to capture short-run transmission dynamics. GFEVD entries are averaged across all $T = 1,423$ time points to produce the static summary in Table II.

The TSI of 41.43% indicates that nearly half of all forecast error variance in the system is attributable to cross-market shocks, confirming substantial financial integration between stablecoin and traditional asset markets and ruling out the view that stablecoin volatility is a purely internal phenomenon.

Three findings emerge:

First, both stablecoins are net transmitters of volatility rather than passive absorbers. USDT and USDC carry NET spillovers of +6.25 and +1.93 respectively, meaning both export more forecast error variance than they receive. This directly challenges the narrative that stablecoins dampen stress: when a stablecoin itself comes under pressure, it becomes a source of systemic disruption, propagating shocks outward to crypto and traditional markets alike - the precise mechanism that rendered the March 2023 SVB episode a system-wide event rather than an isolated pricing anomaly.

Second, the single largest external source of volatility for each stablecoin is the *other* stablecoin. USDT accounts for 13.34% of USDC’s forecast error variance; USDC accounts for 11.62% of USDT’s - both larger contributions than any single crypto or traditional factor. This bilateral linkage means stress in one stablecoin propagates directly to the other, amplifying rather than containing a crisis. It also corroborates the cross-lagged coefficient identified in the regression (USDC Vol_{t−1} transmits to USDT: 0.1359*): that lead-lag relationship reflects a broader pattern of mutual dependence, confirmed here at the system level.

Third, the direction of spillovers between crypto and traditional markets contradicts conventional assumptions. DXY (NET = −5.20), XAU (−5.30), and KBW

TABLE II: The Static Volatility Spillovers (TVP-VAR)

	USDT	USDC	BTC	ETH	DXY	XAU	S&P500	KBW	FROM others
USDT	68.63	11.62	4.68	6.80	2.24	1.90	2.40	1.73	31.37
USDC	13.34	69.50	3.35	5.09	1.72	2.23	2.66	2.12	30.51
BTC	4.62	3.04	52.38	26.85	2.51	4.22	3.70	2.68	47.62
ETH	6.76	5.61	24.46	47.78	2.99	4.15	4.80	3.46	52.23
DXY	2.61	1.98	2.83	3.72	62.11	12.20	8.21	6.33	37.88
XAU	3.74	3.04	4.00	4.73	11.39	59.62	6.95	6.52	40.37
S&P500	3.01	4.66	3.70	4.49	5.99	4.93	54.90	18.32	45.10
KBW	3.54	2.49	3.26	3.75	5.84	5.44	22.02	53.66	46.34
TO others	37.62	32.44	46.28	55.43	32.68	35.07	50.74	41.16	—
NET	+6.25	+1.93	-1.34	+3.20	-5.20	-5.30	+5.64	-5.18	—

Total Spillover Index (TSI) = 41.43%

Notes: TVP-VAR estimated with forgetting factors ($\lambda = 0.99$, $\kappa = 0.96$). GFEVD averaged over 1,423 time points; $H = 10$ -step-ahead forecast horizon; $p = 2$ lags selected by BIC. Rows report volatility *received* (FROM); columns report volatility *transmitted* (TO). NET = TO - FROM; positive values denote net transmitters.

(−5.18) are all net receivers of volatility, while the S&P 500 (+5.64) transmits to the system. Rather than traditional macro variables driving crypto, it is equity market stress and crypto dynamics that spill into currency and commodity markets — a finding with direct implications for how regulators should assess crypto’s systemic footprint.

Reserve-quality regulation under the GENIUS Act marginally reduces the probability that a stablecoin faces redemption pressure, but it does not neutralise the transmission channel that activates once that pressure materialises. **As long as stablecoins remain net transmitters with strong bilateral linkages to one another, a crisis in either instrument will propagate outward regardless of reserve quality.** This suggests that explicit capital buffers, rather than liquidity requirements, are necessary as a mechanism to absorb shocks before they reach the spillover network. The central hypothesis — that reserve-quality regulation lowers the likelihood of stress but does not stop its spread once it starts — is thus strengthened from a market-structure observation to a systemic risk claim by the spillover evidence. Control of propagation would require a completely different tool, which is not mandated by the GENIUS Act.

B. A Balance-Sheet Approach to Peg Resilience

When USDC de-pegged in March 2023, the shock spread outward through the stablecoin channel and into wider markets, which is consistent with USDC acting as a net volatility transmitter rather than a passive absorber. The spillover analysis demonstrates that stablecoin stress is not self-contained. However, the TVP-VAR, which functions at the system level, is unable to address the causal question: why was USDC, and not USDT, the source of that stress? Two instruments that were supposed to be dollar equivalents and were trading on the same exchanges at the same time resulted in

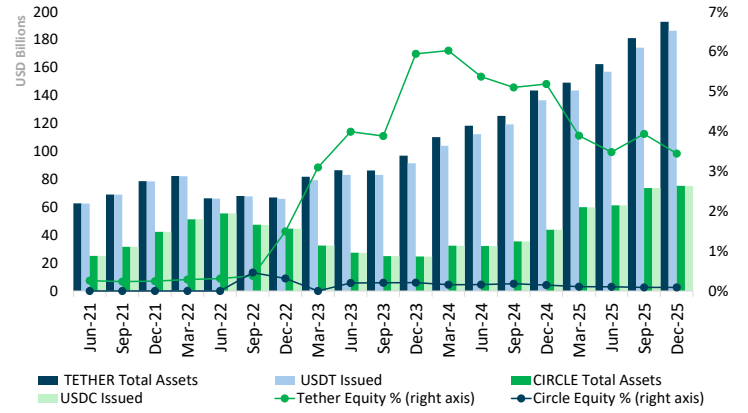


Fig. 4: Evolution of USDC and USDT tokens, reserves and equity buffer of issuers.

wildly divergent market reactions. **The answer cannot lie in reserve quality alone; Circle’s reserves were already broadly GENIUS-compliant in March 2023.** It must lie somewhere in the structural composition of each issuer’s balance sheet. In order to determine the balance-sheet conditions that distinguish peg resilience from peg fragility, we reconstruct the daily mark-to-market reserve portfolios of Circle and Tether. We then test those conditions directly against secondary-market pricing across market regimes, including the SVB crisis window, moving from the market-level evidence of what happened to the issuer-level evidence of why it happened.

Over the past five years, USDC and USDT have followed markedly different paths in terms of growth, reserve design, and perceived issuer credibility. USDT displays sustained expansion in both total assets and outstanding tokens, including through the March 2023 systemic stress episode, and maintains a generally larger equity buffer for most of the sample. In contrast, USDC experienced a sharp contraction in circulating supply and a prolonged stagnation in asset growth following the March 2023 Silicon Valley Bank (SVB) episode, consistent with a confidence shock in U.S.-linked banking infrastructure that triggered redemption pressure and a partial reallocation toward alternative quote currencies. USDC market capitalization begins to recover in 2025, around the period of the GENIUS Act’s approval, while Circle’s reserve mix remains broadly unchanged. (Fig. 4, Fig. 5).

Tether adopts a more diversified, yield-oriented allocation with non-trivial exposure to corporate bonds and other longer-dated assets (including secured loans, metals, and other investments such as digital tokens). On average, these categories represent approximately 17% of reserves, implying higher valuation sensitivity under stress. At the same time, Tether’s cash and deposit share declines materially after 2022, approaching near-zero by 2024, which may reduce its capacity to meet large re-

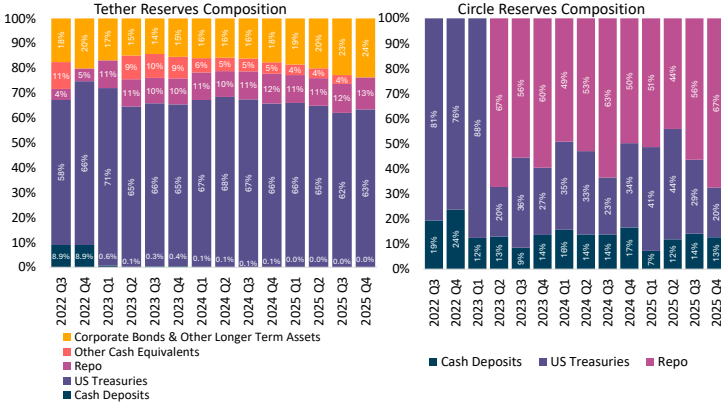


Fig. 5: Comparative Reserve Allocation: Internal Composition of Tether and Circle Backing Assets.

demptions without resorting to asset sales at a discount. Despite this lower immediate cash share, USDT retains a more visible capital buffer (i.e., reserves in excess of tokens outstanding), which can mechanically support confidence under run risk by absorbing valuation shocks before threatening par convertibility.

Circle’s reserve composition, by comparison, becomes increasingly concentrated in very short-maturity instruments after the March 2023 SVB episode, with Repo rapidly becoming the dominant category while direct U.S. Treasury holdings are reduced. By end-2025, Circle maintains a meaningful cash deposit buffer (approximately 13% at year-end), alongside a high share of repo, consistent with a conservative liquidity-first reserve management approach. These patterns are summarized in Fig. 5.

To study issuer resilience under stress and quantify how reserve structure maps into peg dynamics, we reconstruct daily reserve portfolios for Tether (USDT) and Circle (USDC) by interpolating quarterly disclosures with observable market prices and rates. Reserve reports span September 2022 (Granular data available for decomposition to Mark-to-Market effectively) through December 2025 and provide fair value reserve snapshots at quarter-end. This approach moves beyond static disclosure snapshots and yields time-varying measures of equity buffers and liquidity composition, including during the March 2023 crisis window.

We first adjust the aggregate reserve level for daily net issuance/redemptions using a mint/burn proxy inferred from the daily change in market capitalization from Coingecko. Between disclosure dates, we evolve each reserve asset line item using duration-based MTM approximations:

$$Asset_{e,t} = Asset_{e,t-1} - D_{e,t} \cdot (y_{e,t} - y_{e,t-1}) \cdot Asset_{e,t-1} \quad (10)$$

where e indexes the reserve asset line item, $D_{e,t}$ denotes the average duration for that item over the quarter as

reported in the disclosures, and $y_{e,t}$ is the corresponding daily reference rate used to value that line item. For each asset class:

a) *T-Bills*: Treasury holdings are MTM using changes in the 3-month U.S. Treasury yield. Duration varies from 45 to 90 days across the quarter considered.

b) *Repo and Money Market Instruments*: Overnight repo positions are modeled with one-day duration using SOFR rate changes. Reverse repo and money market instruments use maturity-matched short-term funding SOFR rates.

c) *Digital Assets*: For Tether, disclosed Bitcoin and gold holdings are converted into quantities at each disclosure date and MTM daily using BTC and XAU prices. Quantities are assumed constant within each quarter.

d) *Corporate Bonds*: Corporate bond holdings are MTM using Bloomberg BVAL AA yield curves¹¹ with duration assumptions consistent with disclosure reports.

e) *Cash and Deposits*: Cash balances and secured loans are assumed constant between disclosure dates.

All reference rates are obtained from Bloomberg, and all categories are rebalanced to the disclosed fair values at each quarter-end. From the reconstructed balance sheets, we compute the following metrics to add as parameters in our analysis on how issuer balance-sheet strength and liquidity translate into different peg dynamics across coins and exchanges:

- Equity Buffer Ratio: $K_t = \frac{\text{Reserves}_t - \text{Tokens Outstanding}_t}{\text{Reserves}_t}$
- Liquidity Ratio: $L_t = \frac{\text{Cash}_t + \text{Repo}_t}{\text{Total Reserves}_t}$
- T-bill Share: $D_t = \frac{\text{T-Bills}_t}{\text{Total Reserves}_t}$

These parameters aim to link the structural reserve composition to observed cross-currency pricing distortions during the March 2023 stress window. In particular, stablecoins with lower immediate liquidity ratios are expected to experience larger discounts under redemption pressure due to anticipated fire-sale risk.

1) *Regressing Structural Drivers on Peg Deviations*: To quantify issuer-specific pricing dynamics, we estimate panel regressions of daily stablecoin log peg deviations across exchanges for USDT and USDC. Let $y_{e,t} = \log(P_{e,t})$, where $P_{e,t}$ is the USD closing price of the stablecoin on exchange e at date t . Under perfect peg maintenance, $y_{e,t} = 0$; positive (negative) values indicate a premium (discount) relative to par. **Our core hypothesis is that reserve composition and balance-sheet strength affect peg dynamics differently across issuers (USDT vs. USDC), consistent with their distinct roles in crypto markets.**

¹¹The yield curve is constructed daily from USD denominated senior fixed-rate bonds from U.S. companies with ratings AA+, AA and AA-.

We also include exchange indicator dummies to capture persistent, time-invariant differences in venue-level pricing. We report only the statistically significant exchange effects; in economic magnitude they are typically small relative to trading frictions, consistent with limited exploitable arbitrage once fees, slippage, and transfer constraints are accounted for.

Our baseline specification is:

$$y_{e,t} = \alpha + \beta_0 y_{e,t-1} + \beta_1 \text{Safety}_t + \beta_2 \text{VIX}_t + \beta'_3 \mathbf{X}_{e,t-1} + \sum_{j \in \mathcal{E} \setminus \{0\}} \theta_j \mathbb{1}\{e = j\} + \varepsilon_{e,t}, \quad (11)$$

where $\mathbb{1}\{e = j\}$ are exchange dummies (with one omitted baseline exchange), and $\mathbf{X}_{e,t-1}$ includes a lagged high–low spread proxy and lagged log-volume. Standard errors are Newey–West HAC with 7 lags. Exchange effects are absorbed via indicator dummy variables.

- *Reserve-safety measures:* Using the reconstructed balance-sheet metrics from previous section, we include the Liquidity Ratio L_t and Equity Buffer Ratio K_t as issuer fundamentals. For USDT, we include the z-scores $z(L_t)$ and $z(K_t)$ separately because $\text{Corr}(L_t, K_t) = -0.80$, so a single additive index can mask distinct channels. For USDC, we use the composite index:

$$\text{Safety}_t = z(L_t) + z(K_t),$$

where each component is standardized in-sample within each estimation sample to have zero mean and unit variance, which avoids look-ahead bias and makes coefficients comparable across specifications. The dependent variable is winsorized at the 0.1% and 99.9% tails.

- *Regimes:* We report four specifications for each stablecoin: (i) full sample, (ii) calm period (Jul. 2023–Dec. 2025), (iii) high-VIX stress regime using a top-decile threshold ($\text{VIX} \geq 22.83$), and (iv) an SVB-event interaction specification (Mar. 9–20, 2023). In interaction models, VIX is centered before multiplication to reduce multicollinearity.

Table III for USDT shows substantial persistence in daily peg deviations ($\hat{\beta}_0 \approx 0.87$), with a stable fit across specifications ($R^2 \approx 0.762$). Once persistence is included, $z(L_t)$ and $z(K_t)$ are not individually significant in the main specifications. The VIX index enters positively (marginally significant in several columns), consistent with demand for USDT as a crypto settlement asset during risk-off episodes.

In the high-VIX interaction model, the stress dummy is positive and significant, while the $\text{VIX} \times \text{Stress}$ interaction is negative and significant. This pattern is consistent with a two-stage mechanism: an initial flight

TABLE III: USDT Peg-Deviation Regressions

	(1) Full	(2) Calm	(3) Hi-VIX	(4) SVB†
$y_{e,t-1}$	0.870*** (0.081)	0.845*** (0.081)	0.869*** (0.081)	0.869*** (0.081)
$z(L_t)$	−0.0001 (0.0000)	−0.0000 (0.0001)	−0.0001 (0.0000)	−0.0001 (0.0000)
$z(K_t)$	−0.0001 (0.0001)	−0.0000 (0.0001)	−0.0000 (0.0001)	−0.0001 (0.0001)
VIX	2.38e-5*	1.55e-5	5.53e-5*	2.19e-5*
$\log V_{e,t-1}$	−2.8e-5	−7.7e-5	−2.5e-5	−3.7e-5
Crypto.com FE	3.6e-4**	3.4e-4	3.6e-4**	3.7e-4**
Stress _t			3.8e-4***	−0.0081**
$z(L_t) \times \text{Stress}$			−1.2e-4	0.0028**
$z(K_t) \times \text{Stress}$			−1.8e-4	−7.9e-4**
$\text{VIX}^c \times \text{Stress}$			−7.6e-5**	5.0e-4*
Obs.	3,709	2,742	3,709	3,709
R^2	0.762	0.762	0.762	0.762

Notes: Newey–West HAC standard errors (7 lags) in parentheses. Binance.US is the omitted exchange fixed effect.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE IV: USDC Peg-Deviation Regressions

	(1) Full	(2) Calm	(3) Hi-VIX	(4) SVB†
$y_{e,t-1}$	0.532*** (0.086)	0.496*** (0.040)	0.530*** (0.083)	0.459*** (0.077)
Safety _t	1.62e-5** (8.1e-6)	7.14e-6*** (1.5e-6)	3.30e-6** (1.7e-6)	3.20e-6* (1.8e-6)
VIX	−1.46e-6	−2.6e-7	−1.55e-6*	−3.1e-7
$\log V_{e,t-1}$	−9.26e-6***	−6.50e-6***	−9.86e-6***	−8.00e-6***
Kraken FE	−1.47e-5	−1.79e-5***	−1.36e-5	−2.25e-5***
Stress _t			−2.0e-4*	−0.0022
Safety _t × Stress			5.19e-5*	1.0e-4
$\text{VIX}^c \times \text{Stress}$			1.62e-5**	2.0e-4
Obs.	2,460	1,830	2,460	2,460
R^2	0.301	0.362	0.314	0.385

Notes: Newey–West HAC standard errors (7 lags) in parentheses. Coinbase is the omitted exchange fixed effect.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

into USDT raises its premium, followed by premium compression during peak stress as liquidation demand and deleveraging dominate.

The SVB-window interaction specification shows asymmetric pricing of USDT reserve components under issuer-specific stress: $z(L_t) \times \text{Stress}$ is positive and significant, while $z(K_t) \times \text{Stress}$ is negative and significant. This indicates that the market prices immediate liquidity and capital buffer differently when redeemability risk becomes salient.

On the other hand, Table IV for USDC shows that the reserve-safety coefficient is positive and statistically significant across specifications. Economically, higher reserve safety is associated with *smaller discount episodes* (i.e., less negative $y_{e,t}$ and lower magnitude downside de-pegs), rather than implying tighter adherence in an absolute sense on both sides of parity. While the autoregressive term is economically large ($\hat{\beta}_0 \approx 0.50$ – 0.53), it shows substantially weaker persistence than USDT. In contrast to USDT, the level effect of VIX is generally weak, suggesting that USDC pricing is primarily driven by issuer fundamentals rather than broad macro risk sentiment in normal times. Lagged log-volume enters with

a consistently negative and highly significant coefficient, consistent with intensified trading activity coinciding with larger discounts (e.g., redemption-driven selling pressure).

In the high-VIX interaction specification, the safety effect strengthens ($\text{Safety} \times \text{Stress} > 0$) and the centered $\text{VIX} \times \text{Stress}$ term is positive and significant. A natural interpretation is that conditional on stronger reserve quality, USDC benefits from relative safe-haven demand within crypto during broad market stress.

Overall, the panel evidence supports a structural distinction between the two issuers: **USDC peg dynamics are more sensitive to reserve composition and balance-sheet quality, whereas USDT behaves more like a persistent, sentiment-sensitive offshore liquidity instrument.** USDT is driven by sentiment and persistence, while USDC peg dynamics are driven by reserve composition and balance-sheet quality. **Investors distinguish between thin-cushioned and well-capitalized issuers rather than high-quality and low-quality reserves when redeemability risk becomes significant,** as it did during the SVB window. This indicates that the market already prices the capital buffer channel independently from the reserve quality channel. Reserve mandates do not address the latter distinction, but they do address the former.

2) *The “Bank” Stress Testing*: As Basel III’s routine stress testing aims to ensure that banks remain liquid and solvent, we have proposed a similar stress testing approach for stablecoins. Using the daily reserve portfolios of Circle (USDC) and Tether (USDT), we ran three slightly tweaked, yet more stablecoin-appropriate stress scenarios:

- **Scenario A (“Liquidity Run”)**: The March 2023 episode is evidence that a 30% shock in redemptions is a historically plausible tail risk event. Hence, we check if cash and overnight repos can cover 30% withdrawals of the total coins issued. If not, T-bills or Corporate Bonds must be sold in a panic with a 2% fire-sale penalty (slippage).
- **Scenario B (“Flash Crash”)**: Bitcoin has regularly suffered 50% corrections (May 2021, Nov 2022, Feb 2026), gold saw a 15% drop during the 2008 liquidity crisis, and a 300 bps interest rate shock is the standard extreme stress scenario used by the Federal Reserve. Hence, we check if the issuer’s reserves composition can sustain these tail event price and rate movements.
- **Scenario C (“Doomsday Contagion”)**: During the 2008 financial crisis, the default rates went up to 10% on corporate bonds and 25% on secured loans. Replicating the 2008 crisis as a doomsday

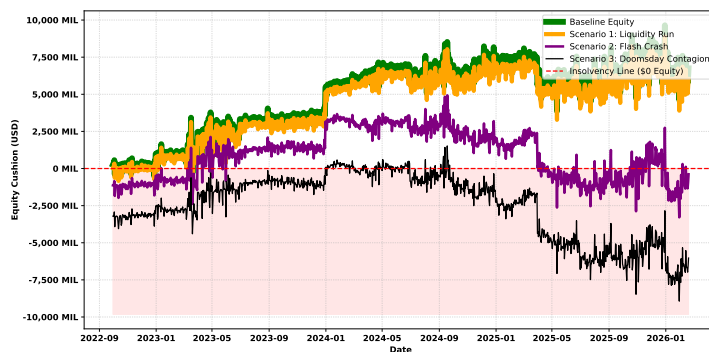


Fig. 6: Tether Stress Test: Equity cushion sensitivity to market and credit contagion shocks.

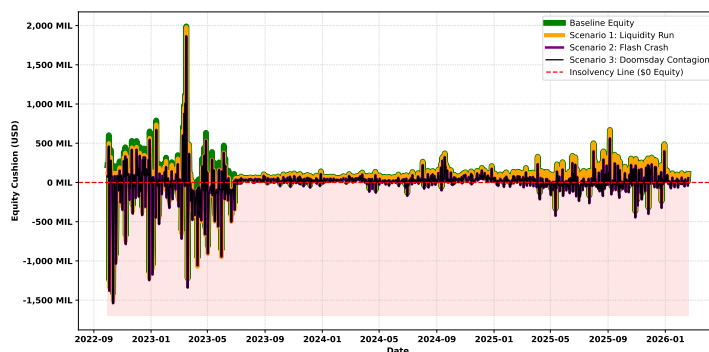


Fig. 7: Circle (USDC) Stress Test: High resilience to market shocks with primary exposure to liquidity-driven fire sales.

scenario, we applied haircuts of 10% on corporate bonds, 25% on secured loans, and 50% on “other investments”, while also combining the first two scenarios.

While one would presume that Tether follows a riskier business model, our stress tests reveal an inverted reality. Tether’s business model (which emphasizes building a capital buffer) is better-suited for the stablecoin business than Circle’s business model (which seems “safer” with risk-free assets).

Tether follows a shadow banking model by taking deposits (USDT in this case) and investing them in higher-yield, riskier, and less transparent assets (Secured Loans, BTC, Corporate Bonds). This model entails procyclical risks. When the market is good, Tether builds a massive surplus. However, as the market crashes, the risky assets would fail. The gap between the green and black line on Fig. 6 implies the existence of market and credit risk hidden underneath the stablecoin. Tether’s solvency under a doomsday scenario depends heavily on the quality of its private credit and “other investment”. If these reach a 25-50% haircut, the surplus equity would evaporate. **However, Tether’s “supposedly riskier” business model performs perfectly well under our liquidity run scenario, which is analogous to the aftermath of the SVB crisis.**

Circle follows a “supposedly safer” narrow banking business model in which it holds 100% of its deposit liabilities in cash equivalents or highly safe T-bills. The non-existent gap between (or the perfectly overlaid) green and black lines on Fig. 7 suggests that Circle does not have any market or credit risk (as it does not hold any risky assets). All of its assets are more liquid in nature than its liabilities. As Circle does not hold BTC or Corporate Bonds, a flash crash or a doomsday scenario has no extra impact on its equity cushion. However, before having to consider these rather extreme stress scenarios, it is important to note that Circle consistently fails during the liquidity run scenario as its equity cushion repeatedly goes negative due to the lack of a buffer. **Circle’s primary risk is not solvency but rather liquidity: it cannot afford to allocate capital in the wrong place (or bank) during a 30% sudden redemption run. During March 2023, Circle’s reserves were held at SVB, exposing USDC to the full impact of the bank’s sudden collapse and triggering an immediate crisis of confidence in the stablecoin’s peg.** Circle satisfied every standard the GENIUS Act mandates yet failed the one stress scenario that actually occurred, while Tether — non-compliant by every conventional metric — passed: the difference was not reserve quality but equity buffer, and that distinction is precisely what the Act does not address.

The GENIUS Act’s omission of mandatory capital buffers and rigorous stress testing limits its effectiveness in ensuring true stablecoin resilience. Consequently, by not requiring issuers to maintain surplus equity to absorb severe market and liquidity shocks, the legislation leaves fundamental structural vulnerabilities unaddressed and fails to fully insulate the ecosystem from de-pegging events.

This completes the paper’s two-part explanation. The system as a whole is affected when a stablecoin fails, as demonstrated by the spillover analysis in Section III-A. The reserve analysis and stress tests here demonstrate that Circle’s failure was due to the lack of excess reserves to withstand a confidence shock before it hit the market, not to the low quality of its reserve assets. The GENIUS Act solves the first issue but not the second, according to the two analyses, which together support the same conclusion from different angles.

IV. CONCLUSION

Stablecoins are widely considered as a store of value that play a stabilizing role during periods of stress. However, the March 2023 SVB episode demonstrated that stablecoins can themselves become a source of dislocation when issuer redeemability is called into

question. Understanding what drives pricing deviations between instruments designed to be equivalent is therefore both empirically and policy-relevant. **The evidence repeatedly suggests that reserve composition is not the primary structural determinant of peg credibility: across our spillover analysis, panel regressions, and stress tests, the issuer with the larger equity buffer consistently demonstrated greater peg resilience, while the issuer with higher-quality but thinner-cushioned reserves remained vulnerable precisely when that liquidity was most needed.**

The GENIUS Act is a meaningful step toward standardization, transparency and oversight of stablecoin issuers. Our findings suggest that the framework is far from complete. While the Act prioritizes asset segregation, it must require issuers to hold adequate loss-absorbing resources and operational capital as emphasized in a joint banking-industry comment letter submitted to the U.S. Department of the Treasury [3]. As recent regulatory frameworks emphasize, holding high-quality liquid assets alongside sufficient capital buffers is essential to cover expected outflows during a stress period and ensure issuers can make customers whole during operational or market failures. **Therefore, policymakers must mandate periodic liquidity stress testing and establish a minimum equity buffer ratio, analogous to the Basel III stress testing and Leverage Ratio requirement.** Our analysis indicates that a (mandated) buffer would provide the necessary loss-absorbing capacity to prevent fire-sale insolvencies and prevent material risks to the broader financial system.

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